

# Network goodness calculus proposition

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Seminar

Part two

# Square distance to network

## Theorem 1.

For a given network  $\bigcup_{j=1}^s \overline{A_j B_j}$ , and for each  $P_i$ ,  $i = 1, \dots, n$ , there exists

$$d_i^2 = d^2(P_i) = \min_{j \leq s} \{d^2(P_i, \overline{A_j B_j})\}, \quad i = 1, \dots, n. \quad (0.1)$$

## Proof of Theorem 1.

Immediately, since the network is finite.

Induction on  $j = 1, \dots, s - 1, s$ :

- For  $j = 1$ :  $d_i^2 = d^2(P_i, \overline{A_1 B_1}) = \min_{j \leq 1} \{d^2(P_i, \overline{A_j B_j})\}.$

- For  $j = 2$ :

$$d_i^2 = \min \left\{ d^2(P_i, \overline{A_2 B_2}), \min_{j \leq 1} \{d^2(P_i, \overline{A_j B_j})\} \right\} = \min_{j \leq 2} \{d^2(P_i, \overline{A_j B_j})\}.$$



## Induction step

For  $j = s - 1$ , suppose that for graph  $\bigcup_{j=1}^{s-1} \overline{A_j B_j}$ ,

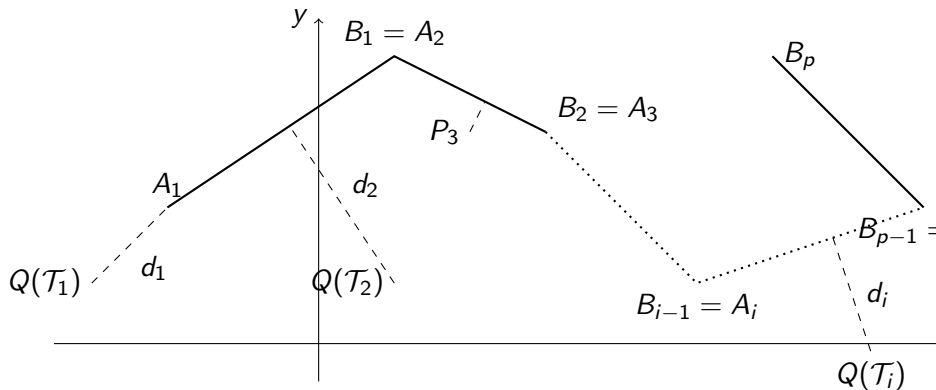
$$d_i^2 = \min_{j \leq s-1} \{d^2(P_i, \overline{A_j B_j})\}.$$

If  $j = s$ , then  $\bigcup_{j=1}^s \overline{A_j B_j} = \bigcup_{j=1}^{s-1} \overline{A_j B_j} \cup \overline{A_s B_s}$ , and

$$\begin{aligned} d_i^2 &= \min \left\{ \min_{j \leq s-1} \{d^2(P_i, \overline{A_j B_j})\}, d_i^2(P_i, \overline{A_s B_s}) \right\} \\ &= \min_{j \leq s} \{d^2(P_i, \overline{A_j B_j})\}, \end{aligned}$$

for each  $i = 1, \dots, n$ . Since all the above, the square distance existence is proven. It is unique by a general minimum definition.

## Demand zone inertia idea



Slika: Calculating the moment zone inertia along a network segment's path.

# Gravitating and attracting network

Distance values given in advance:

- attracting distance  $d$
- gravitating distance  $D$

## Definition 2 (Gravitating and attracting zones).

A demand zone  $\mathcal{T}_i$  gravitates to the network iff  $d_i < D$

A demand zone  $\mathcal{T}_j$  is attracted by the network iff  $d_j < d$

# Network inertia

## Definition 3.

The inertia of the  $i$ -th demand zone:

$$M_i = Q(\mathcal{T}_i) \cdot d_i^2, \quad i = 1, \dots, n \quad (0.2)$$

## Definition 4 (Network inertia and gravitating zones' number).

The network inertia is the sum of all network gravitated zones inertia:

$$M = \sum_{i=1}^k M_i = \sum_{i=1}^k Q(\mathcal{T}_i) d_i^2, \quad (0.3)$$

where  $k$  is the gravitating demand zones' number.